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Java SE 27 & JDK 27

Class Array

java.lang.Object
    java.lang.reflect.Array

```
public final class Array
extends Object
```

The `Array` class provides static methods to dynamically create and access Java arrays.

`Array` permits widening conversions to occur during a get or set operation, but throws an `IllegalArgumentException` if a narrowing conversion would occur.

Since:

1.1

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static Object	**get**(Object array, int index)	Returns the value of the indexed component in the specified array object.
static boolean	**getBoolean**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a `boolean`.
static byte	**getByte**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a `byte`.
static char	**getChar**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a `char`.
static double	**getDouble**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a `double`.

static float	**getFloat**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a float.
static int	**getInt**(Object array, int index)	Returns the value of the indexed component in the specified array object, as an int.
static int	**getLength**(Object array)	Returns the length of the specified array object, as an int.
static long	**getLong**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a long.
static short	**getShort**(Object array, int index)	Returns the value of the indexed component in the specified array object, as a short.
static Object	**newInstance**(Class<? > componentType, int length)	Creates a new array with the specified component type and length.
static Object	**newInstance**(Class<? > componentType, int... dimensions)	Creates a new array with the specified component type and dimensions.
static void	**set**(Object array, int index, Object value)	Sets the value of the indexed component of the specified array object to the specified new value.
static void	**setBoolean** (Object array, int index, boolean z)	Sets the value of the indexed component of the specified array object to the specified boolean value.
static void	**setByte**(Object array, int index, byte b)	Sets the value of the indexed component of

		the specified array object to the specified `byte` value.
`static void`	**`setChar`**`(Object array, int index, char c)`	Sets the value of the indexed component of the specified array object to the specified `char` value.
`static void`	**`setDouble`**`(Object array, int index, double d)`	Sets the value of the indexed component of the specified array object to the specified `double` value.
`static void`	**`setFloat`**`(Object array, int index, float f)`	Sets the value of the indexed component of the specified array object to the specified `float` value.
`static void`	**`setInt`**`(Object array, int index, int i)`	Sets the value of the indexed component of the specified array object to the specified `int` value.
`static void`	**`setLong`**`(Object array, int index, long l)`	Sets the value of the indexed component of the specified array object to the specified `long` value.
`static void`	**`setShort`**`(Object array, int index, short s)`	Sets the value of the indexed component of the specified array object to the specified `short` value.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Details

newInstance

```
public static Object newInstance(Class<?> componentType,
                                 int length)
                          throws NegativeArraySizeException
```

Creates a new array with the specified component type and length.

Implementation Requirements:

Invoking this method is equivalent to creating an array as follows:

```
Array.newInstance(componentType, new int[]{length});
```

The number of dimensions of the new array must not exceed 255.

Parameters:

`componentType` - the `Class` object representing the component type of the new array

`length` - the length of the new array

Returns:

the new array

Throws:

`NullPointerException` - if the specified `componentType` parameter is null

`IllegalArgumentException` - if componentType is `Void.TYPE` or if the number of dimensions of the requested array instance exceed 255.

`NegativeArraySizeException` - if the specified `length` is negative

newInstance

```
public static Object newInstance(Class<?> componentType,
                                 int... dimensions)
                          throws IllegalArgumentException,
                                 NegativeArraySizeException
```

Creates a new array with the specified component type and dimensions. If `componentType` represents a non-array class or interface, the new array has `dimensions.length` dimensions and `componentType` as its component type. If `componentType` represents an array class, the number of dimensions of the

new array is equal to the sum of `dimensions.length` and the number of dimensions of `componentType`. In this case, the component type of the new array is the component type of `componentType`.

The number of dimensions of the new array must not exceed 255.

Parameters:

`componentType` - the `Class` object representing the component type of the new array

`dimensions` - an array of `int` representing the dimensions of the new array

Returns:

the new array

Throws:

`NullPointerException` - if any of the specified `componentType` or `dimensions` arguments is null

`IllegalArgumentException` - if the specified `dimensions` argument is a zero-dimensional array, if componentType is `Void.TYPE`, or if the number of dimensions of the requested array instance exceed 255.

`NegativeArraySizeException` - if any of the components in the specified `dimensions` argument is negative.

getLength

```
public static int getLength(Object array)
                     throws IllegalArgumentException
```

Returns the length of the specified array object, as an `int`.

Parameters:

`array` - the array

Returns:

the length of the array

Throws:

`NullPointerException` - if `array` is `null`

`IllegalArgumentException` - if `array` is not an array

get

```
public static Object get(Object array,
                         int index)
                  throws IllegalArgumentException,
                         ArrayIndexOutOfBoundsException
```

Returns the value of the indexed component in the specified array object. The value is automatically wrapped in an object if it has a primitive type.

Parameters:

`array` - the array

`index` - the index

Returns:

the (possibly wrapped) value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

getBoolean

```
public static boolean getBoolean(Object array,
                                 int index)
                          throws IllegalArgumentException,
                                 ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `boolean`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getByte

```
public static byte getByte(Object array,
                           int index)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `byte`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getChar

```
public static char getChar(Object array,
                           int index)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `char`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getShort

```
public static short getShort(Object array,
                             int index)
                      throws IllegalArgumentException,
                             ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `short`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getInt

```
public static int getInt(Object array,
                         int index)
                  throws IllegalArgumentException,
                         ArrayIndexOutOfBoundsException
```

Returns the value of the indexed component in the specified array object, as an `int`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getLong

```
public static long getLong(Object array,
                           int index)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `long`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getFloat

```
public static float getFloat(Object array,
                             int index)
                      throws IllegalArgumentException,
                             ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `float`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

getDouble

```
public static double getDouble(Object array,
                               int index)
                        throws IllegalArgumentException,
                               ArrayIndexOutOfBoundsException
```

Returns
the value of the indexed component in the specified array object, as a `double`.

Parameters:

`array` - the array

`index` - the index

Returns:

the value of the indexed component in the specified array

Throws:

`NullPointerException` - If the specified object is null

`IllegalArgumentException` - If the specified object is not an array, or if the indexed element cannot be converted to the return type by an identity or widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`get(Object, int)`

set

```
public static void set(Object array,
                       int index,
                       Object value)
                throws IllegalArgumentException,
                       ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified new value. The new value is first automatically unwrapped if the array has a primitive component type.

Parameters:

`array` - the array

`index` - the index into the array

`value` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the array component type is primitive and an unwrapping conversion fails

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

setBoolean

```
public static void setBoolean(Object array,
                              int index,
                              boolean z)
                       throws IllegalArgumentException,
                              ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `boolean` value.

Parameters:

`array` - the array

`index` - the index into the array

`z` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

setByte

```
public static void setByte(Object array,
                           int index,
                           byte b)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `byte` value.

Parameters:

`array` - the array

`index` - the index into the array

`b` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

setChar

```
public static void setChar(Object array,
                           int index,
                           char c)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `char` value.

Parameters:

`array` - the array

`index` - the index into the array

`c` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

setShort

```
public static void setShort(Object array,
                            int index,
                            short s)
                     throws IllegalArgumentException,
                            ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `short` value.

Parameters:

`array` - the array

`index` - the index into the array

`s` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

IllegalArgumentException - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

ArrayIndexOutOfBoundsException - If the specified index argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

set(Object, int, Object)

setInt

```
public static void setInt(Object array,
                          int index,
                          int i)
                   throws IllegalArgumentException,
                          ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified int value.

Parameters:

array - the array

index - the index into the array

i - the new value of the indexed component

Throws:

NullPointerException - If the specified object argument is null

IllegalArgumentException - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

ArrayIndexOutOfBoundsException - If the specified index argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

set(Object, int, Object)

setLong

```
public static void setLong(Object array,
                           int index,
                           long l)
                    throws IllegalArgumentException,
                           ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified long value.

Parameters:

`array` - the array

`index` - the index into the array

`l` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

setFloat

```
public static void setFloat(Object array,
                            int index,
                            float f)
                     throws IllegalArgumentException,
                            ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `float` value.

Parameters:

`array` - the array

`index` - the index into the array

`f` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

setDouble

```
public static void setDouble(Object array,
                             int index,
                             double d)
                      throws IllegalArgumentException,
                             ArrayIndexOutOfBoundsException
```

Sets the value of the indexed component of the specified array object to the specified `double` value.

Parameters:

`array` - the array

`index` - the index into the array

`d` - the new value of the indexed component

Throws:

`NullPointerException` - If the specified object argument is null

`IllegalArgumentException` - If the specified object argument is not an array, or if the specified value cannot be converted to the underlying array's component type by an identity or a primitive widening conversion

`ArrayIndexOutOfBoundsException` - If the specified `index` argument is negative, or if it is greater than or equal to the length of the specified array

See Also:

`set(Object, int, Object)`

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